HyQ-LC: Improving Light-Curve Classification with Hybrid Quantum–Classical Learning

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Light curves reveal stellar variability and support studies of stellar evolution and cosmic distances. Reliable classification is therefore important for large time-domain surveys. We present HyQ-LC for ten-class stellar light-curve classification using 359 handcrafted features and 633,034 training objects. The main model combines a window multilayer perceptron (MLP) backbone with a nine-qubit readout using amplitude encoding and angle re-uploading. Both this hybrid and its pure-MLP control have 44,238 parameters. On a shared 20,000-object test set, the hybrid improves macro-averaged F1 (Macro-F1) by 0.8276 and 1.2393 percentage points on two fine-tuning seeds. A separate fixed-fusion configuration achieves slightly higher Macro-F1 but lower accuracy than approximately budget-matched LightGBM. Its positive class-level differences are concentrated in CEP, RRC, and ROT, while the full-budget LightGBM reference scores higher on both metrics. All circuits are simulated classically. The comparisons retain existing validation-selected checkpoints and use the same test objects, with features, predictions, and recomputation code publicly available. The results identify improvements over the tested controls while leaving the specific contribution of the quantum branch unresolved.

1 Introduction

Time-domain surveys generate large collections of light curves that require scalable classification [1]. Feature-based systems such as ALeRCE [2] and incremental frameworks such as TDLight [3] address this task through complementary classification and data-management strategies. Learned representations provide another route, including ASTROMER embeddings [4], time-series foundation models benchmarked by StarEmbed [5], and structured sequence modelling with large language models [6]. These studies provide context; their different inputs, labels, and evaluation protocols prevent direct score comparisons with the present task.

Hybrid quantum–classical learning combines classical processing with trainable quantum representations [7]. Peters et al. demonstrated quantum-kernel classification of supernova spectral features on a noisy processor [8]. Architecture comparisons on tabular data also motivate studying how quantum branches interact with classical networks [9]. However, expressive circuits alone do not guarantee predictive improvements. Bowles et

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al. showed that classical baselines and entanglement-removal controls can substantially change the interpretation of quantum-learning benchmarks [10].

The distinction between a useful hybrid classifier and a useful quantum branch is central to this evaluation. A hybrid can improve on a dense network because of its feature topology, pre-training, or residual head, even when a classical replacement supplies similar information. Parameter matching controls model size but does not remove these alternatives. Handcrafted light-curve features also define a tabular learning problem, making a strong tree-based reference relevant alongside neural controls.

Here we ask whether a hybrid representation improves stellar light-curve classification under controlled parameter budgets. We distinguish comparisons with a parameter-matched pure MLP from replacements of the quantum branch within a shared network. A separate fixed-fusion configuration supports comparisons with classical fusion and an approximately budget-matched LightGBM. Every reported test comparison uses the same 20,000 objects.

2 Models and Evaluation

2.1 Data and task formulation

We use handcrafted features derived from the LEAVES light-curve dataset [11, 12]. The task contains nine variable-star categories and one non-variable category. The labels are CEP, DSCT, EA, EW, M, Non-var, ROT, RRAB, RRC, and SR; Non-var denotes non-variable objects. Each object has 359 precomputed handcrafted features. The training pool contains 633,034 objects, and the reported test cohort contains 20,000 unique objects. EW and SR together account for 61.89% of the test cohort; CEP has only 45 objects. Class counts and classwise scores are listed in Table 4 and provided with the released data [13].

Preprocessing uses frozen training-derived medians and scales, median replacement of non-finite values, and clipping to $[-10, 10]$. The main models use 105,297 validation objects; the fixed-fusion and leaf-budget LightGBM experiments use 10,000. We retain their existing validation-selected checkpoints and hyperparameters. Test predictions are extracted by exact ID and label matching from saved prediction files. The cohort was previously examined during exploratory development, so it is not an untouched confirmation sample. Separate sealed confirmation data remain unused.

We formulate classification as a map from a preprocessed feature vector $\mathbf{x}_i \in \mathbb{R}^{359}$ to a label $y_i \in \{1, \ldots, C\}$, with $C = 10$. The networks receive engineered features rather than raw sequences of observation times and magnitudes. Consequently, the windows below operate over entries of the feature vector; they do not denote consecutive time intervals in a light curve. This common input representation allows direct comparisons with dense MLPs and gradient-boosted trees.

2.2 Classical backbone and model overview

The main HyQ-LC model combines a classical feature extractor with a global quantum readout (Figure 1a). The backbone is a 23,170-parameter window MLP, adapted from the token-based VQC/FQT architectures studied by Chen and Kuo [9] by using local MLPs in place of quantum blocks. Let $\mathbf{r} = \arctan(\mathbf{x})$ denote the elementwise bounded input. Padding $\mathbf{r}$ with one zero produces 120 three-component tokens. Token MLPs and two stages of six-component window mixing yield a 120-dimensional feature vector and ten base logits:

$$\mathbf{h} = B_\eta(\mathbf{r}) \in \mathbb{R}^{120}, \qquad \mathbf{z}_{\text{base}} = W_{\text{base}}\mathbf{h} + \mathbf{b}_{\text{base}}. \tag{1}$$

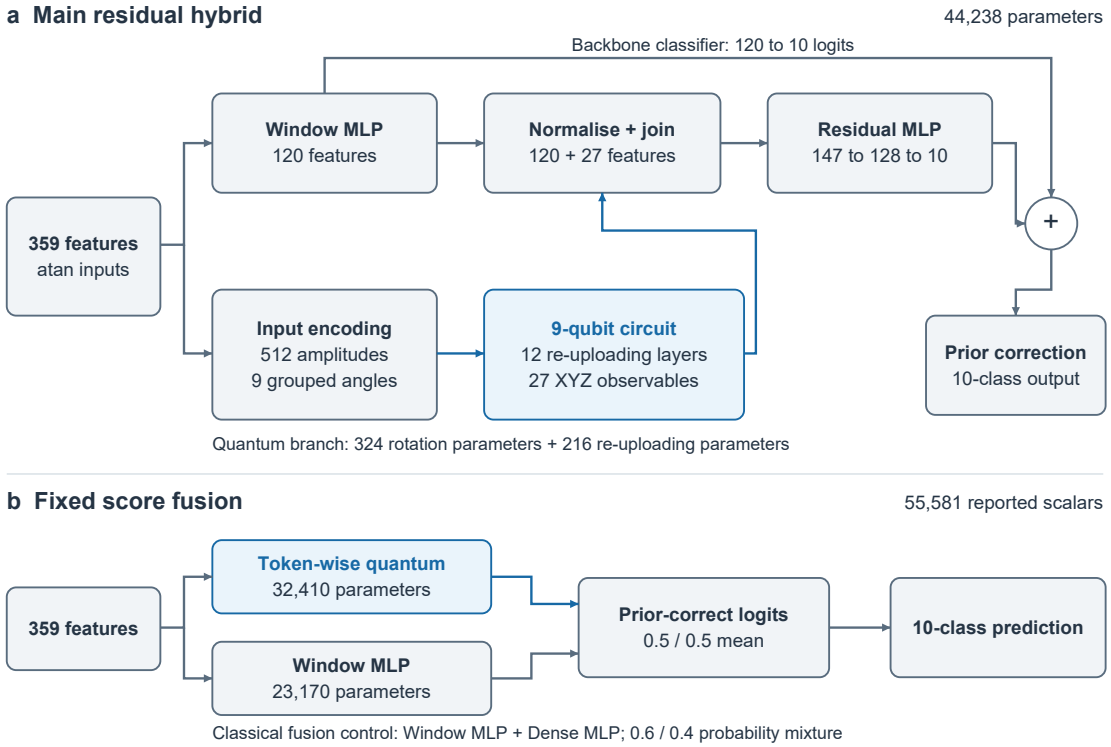


Figure 1: Two distinct hybrid configurations. (a) The main model adds a residual prediction from classical features and a global nine-qubit readout to the backbone logits. The quantum input combines normalised amplitudes and grouped angles. (b) Fixed fusion averages the corrected logits of two independently trained classifiers. Its classical control instead mixes Window MLP and Dense MLP probabilities. The fixed-fusion scalar budgets include one fixed mixing coefficient.

The quoted backbone parameter count includes its classifier. The quantum branch processes all 359 input features in a single nine-qubit register, and its measured features enter a residual classifier alongside $\mathbf{h}$. Retaining $\mathbf{z}_{\text{base}}$ gives the added branch a direct classical prediction to refine. It also makes the complete-model comparison distinct from a test of the quantum readout alone.

2.3 Quantum encoding, circuit, and measurement

The quantum input combines amplitude encoding with grouped angle re-uploading. Define $\bar{r}_k = r_k$ for $0 \le k < 359$ and $\bar{r}_k = 0$ for $359 \le k < 512$, and let $\bar{\mathbf{r}}$ collect these 512 entries. Since $2^9 = 512$, the encoded state is

$$|\psi_0(\mathbf{x})\rangle = \begin{cases} \displaystyle\sum_{k=0}^{511} \frac{\bar{r}_k}{\|\bar{\mathbf{r}}\|_2} |k\rangle, & \|\bar{\mathbf{r}}\|_2 > 10^{-8}, \\ |0\rangle^{\otimes 9}, & \text{otherwise.} \end{cases} \tag{2}$$

This encoding uses no learned projection. Normalisation preserves the direction of the transformed feature vector in the amplitudes. The circuit also receives nine angles formed from contiguous groups of 40 padded entries,

$$\varphi_j(\mathbf{x}) = \frac{2}{40} \sum_{t=0}^{39} \bar{r}_{40j+t}, \qquad j = 0, \ldots, 8. \tag{3}$$

Thus, the grouped angles provide a second input path in addition to the normalised amplitudes. Data re-uploading repeatedly introduces these angles into the trainable circuit [14].

The readout has twelve layers. In layer ℓ, each qubit receives an Euler rotation, followed by a ring of CNOT gates and a data-dependent R_z rotation:

$$|\psi_\ell\rangle = D_\ell(\boldsymbol{\varphi})\, E\, R_\ell |\psi_{\ell-1}\rangle, \qquad \ell = 1, \ldots, 12, \tag{4}$$
$$R_\ell = \prod_{j=0}^{8} R_{z,j}(\theta_{\ell j 3}) R_{y,j}(\theta_{\ell j 2}) R_{z,j}(\theta_{\ell j 1}),$$
$$D_\ell(\boldsymbol{\varphi}) = \prod_{j=0}^{8} R_{z,j}(\omega_{\ell j}\varphi_j + b_{\ell j}).$$

Here $R_{\nu,j}(t) = \exp(-it\sigma_{\nu,j}/2)$ acts on qubit j. The entangler E applies $\mathrm{CNOT}_{j\to(j+1) \bmod 9}$ in order $j = 0, \ldots, 8$, and qubit zero is the least significant basis bit. The trainable θ, ω, and b tensors contain $12 \times 9 \times 3 = 324$, $12 \times 9 = 108$, and 108 parameters, respectively. The complete quantum readout therefore has 540 parameters. Its frequency parameters ω start at one and its shifts b at zero.

The final state is converted to classical features through single-qubit Pauli expectations. Writing $\langle \sigma_j \rangle = \langle \psi_{12} | \sigma_j | \psi_{12} \rangle$, we use

$$\mathbf{q}(\mathbf{x}) = \big(\langle X_0\rangle, \ldots, \langle X_8\rangle, \langle Y_0\rangle, \ldots, \langle Y_8\rangle, \langle Z_0\rangle, \ldots, \langle Z_8\rangle\big)^\top \in \mathbb{R}^{27}. \tag{5}$$

These bounded observables provide a compact representation for the classical classifier [15]. Input-dependent phases and the entangling gates change the measured features jointly; the classification experiments test whether this particular representation is useful. Encoding affects the accessible function class [16], but circuit depth and expressivity alone do not establish predictive performance [17]. All expectations are computed by classical state-vector simulation, and gradients are obtained by automatic differentiation through the simulated circuit.

2.4 Residual fusion and matched classical controls

The classical and quantum features are separately layer-normalised and concatenated. With GELU denoting the hidden activation, the final logits are

$$\mathbf{u} = \big[\mathrm{LN}_c(\mathbf{h}); \mathrm{LN}_q(\mathbf{q})\big] \in \mathbb{R}^{147},$$
$$\mathbf{z}_{\mathrm{hyb}} = \mathbf{z}_{\mathrm{base}} + W_2\, \mathrm{GELU}(W_1\mathbf{u} + \mathbf{b}_1) + \mathbf{b}_2. \tag{6}$$

Here $W_1 \in \mathbb{R}^{128\times147}$ and $W_2 \in \mathbb{R}^{10\times128}$. The final residual layer is initialised to zero, so the added branch initially leaves the backbone logits unchanged. The two normalisation layers and residual MLP contain 20,528 parameters. Together with the backbone and quantum readout, they give $23{,}170 + 540 + 20{,}528 = 44{,}238$ parameters.

The pure-MLP control uses the same bounded input, hidden widths 73 and 207, LayerNorm after each hidden linear layer, GELU activations, and a ten-class output. Including biases and normalisation parameters, its total is also 44,238. This comparison asks whether the full hybrid configuration improves performance at a fixed parameter count.

A second control targets the readout itself while preserving the residual architecture in Eq. (6). Its classical input concatenates the first 359 normalised amplitudes from Eq. (2) and the nine grouped angles. A fixed sign projection $P \in \{-1, +1\}^{368\times26}$ maps these values to 26 components, followed by

$$\mathbf{q}_{\mathrm{MLP}} = A_2\, \mathrm{GELU}(A_1\mathbf{t} + \mathbf{a}_1), \quad \mathbf{t} = P^\top[\mathbf{a}_{0:359}; \boldsymbol{\varphi}]. \tag{7}$$

The vector $\mathbf{a}_{0:359}$ denotes the first 359 state amplitudes, including the fallback in Eq. (2). With $A_1 \in \mathbb{R}^{10\times 26}$, $\mathbf{a}_1 \in \mathbb{R}^{10}$, and a bias-free $A_2 \in \mathbb{R}^{27\times 10}$, the readout has $260+10+270 = 540$ learned parameters. The projection is fixed and contributes no learned parameters. The backbone, normalisation layers, fusion-head initialisation, full parameter count, optimiser-group allocation, and sample order match the quantum configuration. This readout replacement is a more specific test than the complete-model comparison with a pure MLP.

2.5 Fixed score fusion and tree controls

A separate fixed-fusion configuration combines a 32,410-parameter token-wise quantum classifier with the 23,170-parameter window MLP (Figure 1b). The quantum classifier adapts the local three-feature VQC construction of Chen and Kuo [9] to this task. It is a different configuration from the global nine-qubit residual readout. Let $\tilde{\mathbf{z}}^{(k)}$ denote the class-corrected logits of classifier k, as defined below. The quantum fusion uses

$$\mathbf{z}_{\text{fixQ}} = \tfrac{1}{2}\tilde{\mathbf{z}}^{(\text{tokenQ})} + \tfrac{1}{2}\tilde{\mathbf{z}}^{(\text{window})}. \tag{8}$$

The classical fusion control combines the window MLP with a 31,460-parameter Dense MLP at the probability level:

$$\mathbf{p}_{\text{fixC}} = 0.6\,\text{softmax}(\tilde{\mathbf{z}}^{(\text{window})}) + 0.4\,\text{softmax}(\tilde{\mathbf{z}}^{(\text{dense})}). \tag{9}$$

Predictions select the largest fused logit or probability, respectively. Both fusion rules were selected on validation. Reported budgets are 55,581 and 54,631 scalars, including one fixed mixing coefficient each; each constituent network receives 40 training epochs. Because both the fusion rule and network sizes differ, this is an approximate-budget ensemble comparison.

The LightGBM [18] control targets the fixed hybrid's scalar budget using cumulative leaf outputs across all class trees. For C classes, T boosting rounds, and L leaves per tree, the nominal budget is CTL. The selected model has $C = 10$, $T = 370$, and $L = 15$, giving 3,700 class trees and 55,500 leaf values. A separate full-budget reference uses $T = 362$ and $L = 31$, with 112,220 leaf values. Both retain their validation-selected settings. Leaf counts exclude split-node features and thresholds, so they provide a capacity proxy without equating tree representations, neural parameters, or runtime.

2.6 Training objective and validation selection

Class imbalance is addressed during neural-network training with inverse-frequency class weights. If n_c is the number of training objects in class c and N_{tr} is the training-pool size, set $w_c = N_{\text{tr}}/(Cn_c)$. For a minibatch $\mathcal{B}$, the weighted cross-entropy is

$$\mathcal{L}_{\mathcal{B}} = -\frac{\sum_{i\in\mathcal{B}} w_{y_i} \log\big[\text{softmax}(\mathbf{z}(\mathbf{x}_i))\big]_{y_i}}{\sum_{i\in\mathcal{B}} w_{y_i}}. \tag{10}$$

The weights are computed from training labels only. At validation and inference, logits are adjusted by a separately selected correction strength:

$$\tilde{z}_c = z_c - \alpha \log w_c, \qquad \hat{y} = \arg\max_c \tilde{z}_c, \quad \alpha \in \{0, 0.25, 0.5, 0.75, 1\}. \tag{11}$$

This correction controls the decision shift associated with class-weighted training. Checkpoints and α are selected by validation Macro-F1. For each pure-MLP seed, the learning rate is selected between 10^{-3} and 3×10^{-3} using validation only. ACC in the manuscript tables is evaluated at the same Macro-F1-selected checkpoint.

The pure MLP starts from scratch and receives an 80-epoch budget. The hybrid uses 40 backbone pre-training epochs followed by a 40-epoch branch-training stage. During the first ten branch-training epochs, the backbone learning rate is zero. The selected primary hybrid checkpoints occur at epochs two and four, so their backbone remains at its pre-trained values. The two primary fine-tuning seeds, 42 and 44, vary added weights and sample order while sharing that pre-trained backbone. Thus, the maximum number of training-object presentations is matched, while initialisation, training trajectory, and computational cost differ. The released implementation references specify the optimiser parameter groups and schedules.

2.7 Evaluation metrics and paired uncertainty

For each class c, let TP_c, FP_c, and FN_c be true-positive, false-positive, and false-negative counts on the shared test cohort. The two reported metrics are

$$\mathrm{Macro\text{-}F1} = \frac{1}{C}\sum_{c=1}^{C}\frac{2\mathrm{TP}_c}{2\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}, \qquad \mathrm{ACC} = \frac{\sum_{c=1}^{C}\mathrm{TP}_c}{N_{\mathrm{test}}}, \tag{12}$$

with $N_{\mathrm{test}} = 20{,}000$. Macro-F1 weights every class equally, whereas ACC weights every object equally. They therefore address different aspects of classification on the imbalanced cohort. Table values are percentages, and hybrid-minus-control differences use percentage points.

Uncertainty is evaluated with 10,000 ordinary paired-object bootstrap resamples. For metric M, the difference in resample b is

$$\Delta_M^{*(b)} = 100\left[M(\mathbf{y}_{I_b}, \hat{\mathbf{y}}_{I_b}^{Q}) - M(\mathbf{y}_{I_b}, \hat{\mathbf{y}}_{I_b}^{C})\right], \tag{13}$$

where I_b contains N_{test} indices sampled with replacement. The same indices are used for both predictions, preserving object-level pairing; class proportions are allowed to fluctuate. The 2.5th and 97.5th percentiles give the reported 95% intervals. These intervals condition on the fitted models and exclude training-seed and model-search variation; no multiplicity correction is applied. The release also includes additional seeded runs and numerical ablations.

3 Results

3.1 Comparison with the parameter-matched pure MLP

The residual hybrid had higher Macro-F1 and accuracy than the pure MLP on both primary seeds (Table 1). The F1 interval included zero for seed 42 and was positive for seed 44; both ACC intervals were positive. These differences compare the complete trained configurations, including their different architectures and pre-training histories.

3.2 Replacing the quantum branch

Within the shared residual architecture, the quantum readout had slightly higher F1 point estimates than the matched classical readout (Table 2). Both intervals included zero. This test therefore leaves the predictive contribution specific to the quantum branch unresolved.

Table 1: Parameter-matched comparison on 20,000 test objects. Both models have 44,238 parameters. F1 scores are percentages; differences are in pp and use unrounded scores. Brackets give paired-bootstrap 95% confidence intervals.

Seed	Hybrid F1	MLP F1	ΔF1 (pp)	F1 95% interval	ΔACC (pp), 95% interval
42	92.2913	91.4637	+0.8276	[−0.0650, +1.6945]	+0.6000 [+0.3450, +0.8550]
44	92.3005	91.0611	+1.2393	[+0.5153, +2.0053]	+0.8450 [+0.5900, +1.1000]

Table 2: Readout replacement within the residual hybrid on the same test objects. Both readouts have 540 parameters and both full models have 44,238 parameters. Scores are Macro-F1 (%); differences and intervals are in pp.

Seed	Quantum readout	MLP readout	ΔF1 (pp)	95% interval
42	92.2913	92.2708	+0.0205	[−0.3299, +0.3726]
44	92.3005	92.0039	+0.2965	[−0.0333, +0.6982]

3.3 Fixed fusion and tree controls

The fixed quantum fusion achieved 92.3371% Macro-F1 and 94.7450% ACC (Table 3). It had a small F1 point gain over classical fusion, with essentially unchanged ACC. Against leaf-budget-matched LightGBM, its F1 was 0.1166 pp higher but ACC was 0.1200 pp lower, corresponding to 24 fewer correct classifications. The paired F1 intervals spanned zero for both controls. The full-budget LightGBM reference scored higher on both metrics.

The classwise comparison (Table 4) shows the largest positive F1 differences against the budget-matched tree in CEP (+1.0040 pp), RRC (+1.1429 pp), and ROT (+0.5128 pp). CEP and RRC contain only 45 and 357 test objects. Equal class weighting allows these gains to raise Macro-F1 even when total correct predictions decrease. This is a class-specific pattern, not an improvement across all minority classes.

Table 3: Controls for fixed quantum fusion on the same 20,000 test objects. The hybrid reference scores F1/ACC of 92.3371/94.7450%. Control scores are percentages; hybrid-minus-control differences and paired 95% intervals are in pp.

Control	F1/ACC (%)	ΔF1 (pp), 95% interval	ΔACC (pp)
Classical fusion	92.1510 / 94.7500	+0.1860 [−0.2869, +0.6740]	−0.0050
Budget-matched LightGBM	92.2204 / 94.8650	+0.1166 [−0.8111, +1.0763]	−0.1200
Full-budget LightGBM	92.9681 / 95.1100	−0.6311 [−1.4192, +0.1197]	−0.3650

Table 4: Class distribution and classwise F1 on the same 20,000 test objects. Quantum fusion denotes the fixed-fusion configuration; LightGBM is the 55,500-leaf-value control. Shares and F1 scores are percentages; ΔF1 is quantum fusion minus LightGBM in pp. The last row gives total counts and unweighted mean F1.

Class	Objects	Share (%)	Quantum F1	LightGBM F1	ΔF1 (pp)
CEP	45	0.225	84.3373	83.3333	+1.0040
DSCT	337	1.685	86.3636	86.4536	−0.0899
EA	912	4.560	96.8097	97.1239	−0.3142
EW	6,307	31.535	94.7460	94.7427	+0.0034
M	352	1.760	97.5955	97.5887	+0.0068
Non-var	2,566	12.830	96.0356	96.7356	−0.7000
ROT	2,257	11.285	86.3818	85.8691	+0.5128
RRAB	796	3.980	97.5579	97.7358	−0.1779
RRC	357	1.785	86.0908	84.9478	+1.1429
SR	6,071	30.355	97.4523	97.6740	−0.2217
Total / Macro-F1	20,000	100.000	92.3371	92.2204	+0.1166

4 Discussion and Conclusion

The comparisons separate performance of the complete classifier from evidence about its quantum component. The residual hybrid improves on the tested parameter-matched pure MLP, yet its shared-architecture readout replacements leave the quantum-specific increment unresolved. Together, these results support the complete configuration as a useful candidate while leaving architecture, pre-training, and branch representation as competing explanations for its gains. This distinction follows the benchmark concerns raised by Bowles et al. [10] and complements architecture-level comparisons such as Chen and Kuo [9].

The classwise results explain why model ranking depends on the evaluation criterion. Under Eq. (12), the F1 score of CEP receives the same weight as that of EW despite their very different sample sizes. The fixed hybrid's improvements in CEP, RRC, and ROT can therefore offset declines in other classes at the Macro-F1 level while total correct predictions decrease. This makes the result relevant to class-balanced classification, but it does not establish a uniform improvement for rare objects. In particular, the CEP estimate rests on only 45 test objects, and classwise F1 changes combine precision and recall rather than measuring recovered objects alone.

The tree comparisons further limit what can be inferred from a scalar budget. Similar leaf-output and neural-parameter counts do not imply equivalent function classes, optimisation effort, or inference cost. The full-budget LightGBM reference scores higher on both metrics, showing that the small F1 advantage over its budget-matched counterpart does not persist across tree budgets. Likewise, the different logit and probability mixtures in Eqs. (8) and (9) prevent attributing their ensemble difference solely to the quantum constituent.

The remaining questions concern generalisation and the source of the residual gain. Evaluation on previously unexamined objects would test persistence beyond the explored cohort. Independently pre-trained backbones would distinguish fine-tuning variation from full training variation, while matched fusion rules would sharpen the ensemble comparison. These tests would extend the current evidence rather than change the reported conditional comparisons. All circuits here are classically simulated; neither a quantum-hardware speed advantage nor a general quantum advantage follows from the present results.

5 Data and Code Availability

The repository provides the shared test features, labels, per-object predictions, result tables, and a portable metric-recomputation script [13]. Checkpoint and source-prediction hashes document provenance. Standalone quantum-readout code and original training implementation references are included; the latter require the authors' cluster environment. The release supports independent recomputation of the reported statistics from saved predictions.

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